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span: bounds-safe views for sequences of objects

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Changelog

Changes from R0

- Changed the name of the type being proposed from *array_view* to *span* following feedback from LEWG at the Kona meeting.
- Removed multidimensional aspects from the proposal. *span* is now always single-dimension and contiguous.
- Added details on potential interoperation with the multidimensional *view* type from P0009 [5].
- Removed functions to convert from *span<byte>* to *span<T>* as they are not compatible with type aliasing rules.
- Introduced dependency on P0257 [6] for definition of *byte* type, in order to support *span* as a method of accessing object representation.
- Added section containing proposed wording for inclusion in the standard.
- Simplified *span* interface based on reviewer feedback.

Introduction

This paper presents a design for a fundamental vocabulary type *span*.

The *span* type is an abstraction that provides a view over a contiguous sequence of objects, the storage of which is owned by some other object. The design for *span* presented here provides bounds-safety guarantees through a combination of compile-time and (configurable) run-time constraints.

The design of the *span* type discussed in this paper is related to the *span* previously proposed in N3851 [1] and also draws on ideas in the *array_ref* and *string_ref* classes proposed in N3334 [2]. *span* is closely related to the generalized, multidimensional memory-access abstraction *view* described in P0009 [5]. The *span* proposed here is sufficiently compatible with *view* that interoperability between the two types would be simple and well-defined.

While *view* is proposed by P0009 [5] as a generalized and highly configurable view type that can address needs for specialized domains such as scientific computing, *span* is proposed as a simple solution to the common need for a single-dimensional view over contiguous storage.

Motivation and Scope

The evolution of the standard library has demonstrated that it is possible to design and implement abstractions in Standard C++ that improve the reliability of C++ programs without sacrificing either performance or portability. This proposal identifies a new “vocabulary type” for inclusion in the standard library that enables both high performance and bounds-safe access to contiguous sequences of elements. This type would also improve modularity, composability, and reuse by decoupling accesses to array data from the specific container types used to store that data.

These characteristics lead to higher quality programs. Some of the bounds and type safety constraints of *span* directly support “correct-by-construction” programming methodology – where errors simply do not compile. One of the major advantages of *span* over the common idiom of a “pointer plus length” pair of parameters is that it provides clearer semantics hints to analysis tools looking to help detect and prevent defects early in a software development cycle.

Impact on the Standard

This proposal is a pure library extension. It does not require any changes to standard classes, functions, or headers. It would be enhanced if could depends on the *byte* type and changes to type aliasing behavior proposed in P0257 [6].

However – if adopted – it may be useful to overload some standard library functions for this new type (an example would be *copy()*).

span has been implemented in standard C++ (C++11) and is being successfully used within a commercial static analysis tool for C++ code as well as commercial office productivity software. An open source, reference implementation is available at <https://github.com/Microsoft/GSL> [3].

Design Decisions

View not container

span is simply a view over another object’s contiguous storage – but unlike *array* or *vector* it does not “own” the elements that are accessible through its interface. An important observation arises from this: *span* never performs any free store allocations.

While *span* is a view, it is not an iterator. You cannot perform increment or decrement operations on it, nor dereference it.

No configurable view properties

In the related *view* type described in P0009 [5], properties are used to control policies such as memory layout (column-major, row-major) and location (on heterogenous memory architectures) for specific specializations of *view*. *span* does not require properties as it is always a simple view over contiguous storage. Its memory layout and access characteristics are equivalent to those of a built-in array. This difference should not prevent conversions between *view* and *span* instances, it merely constrains that they could only be available in cases where *view* properties are compatible with the characteristics of *span*.

View length and measurement

The general usage protocol of the *span* class template supports both static-size (fixed at compile time) and dynamic-size (provided at runtime) views. The *Extent* template parameter to *span* is used to provide the extent of the *span*.

```
constexpr ptrdiff_t dynamic_extent = -1;
```

The default value for *Extent* is *dynamic_extent*: a unique value outside the normal range of lengths (0 to *PTRDIFF_MAX* inclusive) reserved to indicate that the length of the sequence is only known at runtime and must be stored within the *span*. A dynamic-size *span* is, conceptually, just a pointer and size field (this is not an implementation requirement, however).

```
int* somePointer = new int[someLength];  
  
// Declaring a dynamic-size span
```

```
// s will have a dynamic-size specified by someLength at construction
span<int> s = { somePointer, someLength };
```

The type used for measuring and indexing into *span* is *ptrdiff_t*. Using a signed index type helps avoid common mistakes that come from implicit signed to unsigned integer conversions when users employ integer literals (which are nearly always signed). The use of *ptrdiff_t* is natural as it is the type used for pointer arithmetic and array indexing – two operations that *span* explicitly aims to replace but that an implementation of *span* would likely rely upon.

A fixed-size *span* provides a value for *Extent* that is between 0 and PTRDIFF_MAX (inclusive). A fixed-size *span* requires no storage size overhead beyond a single pointer – using the type system to carry the fixed-length information. This allows *span* to be an extremely efficient type to use for access to fixed-length buffers.

```
int arr[10];

// deduction of size from arrays means that span size is always correct
span<int, 10> s2 = arr; // fixed-size span of 10 ints
span<int, 20> s3 = arr; // error: will fail compilation
span<int> s4 = arr; // dynamic-size span of 10 ints
```

Value Type Semantics

span is designed as a value type – it is expected to be cheap to construct, copy, move, and use. Users are encouraged to use it as a pass-by-value parameter type wherever they would have passed a pointer by value or a container type by reference, such as *array* or *vector*.

Conceptually, *span* is simply a pointer to some storage and a count of the elements accessible via that pointer. Those two values within a *span* can only be set via construction or assignment (i.e. all member functions other than constructors and assignment operators are *const*). This property makes it easy for users to reason about the values of a *span* through the course of a function body.

These value type characteristics also help provide compiler implementations with considerable scope for optimizing the use of *span* within programs. For example, *span* has a trivial destructor, so common ABI conventions allow it to be passed in registers.

Range-checking and bounds-safety

All accesses to the data encapsulated by a *span* are conceptually range-checked to ensure they remain within the bounds of the *span*. What actually happens as the result of a failure to meet *span*'s bounds-safety constraints at runtime is undefined behavior. However, it should be considered effectively fatal to a program's ability to continue reliable execution. This is a critical aspect of *span*'s design, and allows users to rely on the guarantee that as long as a sequence is accessed via a correctly initialized *span*, then its bounds cannot be overrun.

As an example, in the current reference implementation, violating a range-check results by default in a call to *terminate()* but can also be configured via build-time mechanisms to continue execution (albeit with undefined behavior from that point on).

Conversion between fixed-size and dynamic-size *span* objects is allowed, but with strict constraints that ensure bounds-safety is always preserved. At least two of these cases can be checked statically by leveraging the type system. In each case, the following rules assume the element types of the *span* objects are compatible for assignment.

1. A fixed-size *span* may be constructed or assigned from another fixed-size *span* of equal length.
2. A dynamic-size *span* may always be constructed or assigned from a fixed-size *span*.
3. A fixed-size *span* may always be constructed or assigned from a dynamic-size *span*. Undefined behavior will result if the construction or assignment is not bounds-safe. In the reference implementation, for example, this is achieved via a runtime check that results in *terminate()* on failure.

Element types and conversions

span must be configured with its element type via the template parameter *ValueType*, which is required to be a complete object type that is not an abstract class type. *span* supports either read-only or mutable access to the sequence it encapsulates. To access read-only data, the user can declare a *span<const T>*, and access to mutable data would use a *span<T>*.

Construction or assignment between *span* objects with different element types is allowed whenever it can be determined statically that the element types are exactly storage-size equivalent (so there is no difference in the extent of memory being accessed), and that the types can legally be aliased.

As a result of these rules, it is always possible to convert from a *span<T>* to a *span<const T>*. It is not allowed to convert in the opposite direction, from *span<const T>* to *span<T>*. This property is extremely convenient for calling functions that take *span* parameters.

Element access and iteration

span's interface for accessing elements is largely similar to that of *array*. It overloads *operator[]* for element access, and offers random access iterators, making it adoptable with a minimum of source changes in code that previously used an array, an *array* object, or a pointer to access more than one object. *span* also overloads *operator()* for element access, to provide compatibility with code written to operate against *view*.

span provides both const and mutable random-access iterators over its data, just like *vector* and *array*. All accesses to elements made through these iterators are range-checked (subject to configuration as previously described), just as if they had been performed via the subscript operator on *span*.

```
// [span.elem], span element access
constexpr reference operator[](index_type idx) const;
constexpr reference operator()(index_type idx) const;
constexpr pointer data() const noexcept;

// [span.iter], span iterator support
iterator begin() const noexcept;
iterator end() const noexcept;

const_iterator cbegin() const noexcept;
```

```

const_iterator cend() const noexcept;

reverse_iterator rbegin() const noexcept;
reverse_iterator rend() const noexcept;

const_reverse_iterator crbegin() const noexcept;
const_reverse_iterator crend() const noexcept;

```

Construction

The *span* class is expected to become a frequently used vocabulary type in function interfaces (as a safer replacement of “(pointer, length)” idioms), as it specifies a minimal set of requirements for safely accessing a sequence of objects and decouples a function that needs to access a sequence from the details of the storage that holds such elements.

To simplify use of *span* as a simple parameter, *span* offers a number of constructors for common container types that store contiguous sequences of elements. A summarized extract from the specification illustrates this:

```

// [span.cons], span constructors, copy, assignment, and destructor
constexpr span();
constexpr span(nullptr_t);
constexpr span(pointer ptr, index_type count);
constexpr span(pointer firstElem, pointer lastElem);
template <size_t N>
    constexpr span(element_type (&arr)[N]);
template <size_t N>
    constexpr span(array<remove_const_t<element_type>, N>& arr);
template <size_t N>
    constexpr span(const array<remove_const_t<element_type>, N>& arr);
template <class Container>
    constexpr span(Container& cont);
template <class Container>
    span(const Container&&) = delete;
constexpr span(const span& other) noexcept = default;
constexpr span(span&& other) noexcept = default;
template <class OtherElementType, ptrdiff_t OtherExtent>
    constexpr span(const span<OtherElementType, OtherExtent>& other);
template <class OtherElementType, ptrdiff_t OtherExtent>
    constexpr span(span<OtherElementType, OtherExtent>&& other);

```

To avoid being mistakenly treated as a container – which would lead to invalid memory accesses – *span* deletes rvalue reference constructors that would bind to an expiring temporary object and requires callers to provide explicit containers to bind the *span* against.

It is allowed to construct a span from the null pointer, and this creates an object with *.size() == 0*. Any attempt to construct a span with a null pointer value and a non-zero length is considered a range-check error.

Byte representations and conversions

span depends upon a distinct “byte” type that represents a single addressable byte on any system, for object representation – in preference to common practice of using character types for this purpose. Such a type is defined in the standard library as:

```
enum class byte : unsigned char {};
```

For more details on the proposed byte type please refer to P0257 [6].

A span of any element type that is a standard-layout type can be converted to a `span<const byte>` or a `span<byte>` via the free functions `as_bytes()` and `as_writable_bytes()` respectively. These operations are considered useful for systems programming where byte-oriented access for serialization and data transmission is essential.

```
// [span.objectrep], views of object representation
template <class ElementType, ptrdiff_t Extent>
constexpr span<const byte, ((Extent == dynamic_extent) ? dynamic_extent :
(sizeof(ElementType) * Extent))> as_bytes(span<ElementType, Extent> s)
noexcept;

template <class ElementType, ptrdiff_t Extent>
constexpr span<byte, ((Extent == dynamic_extent) ? dynamic_extent :
(sizeof(ElementType) * Extent))> as_writable_bytes(span<ElementType, Extent>
) noexcept;
```

These byte-representation conversions still preserve const-correctness, however. It is not possible to convert from a `span<const T>` to a `span<byte>` (through SFINAE overload restriction).

Comparisons

span supports all the same comparison operations as a sequential standard library container: element-wise comparison and a total ordering by lexicographical comparison. This helps make it an effective replacement for existing uses of sequential contiguous container types like *array* or *vector*.

```
// [span.comparison], span comparison operators
template <class ElementType, ptrdiff_t Extent>
constexpr bool operator==(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
constexpr bool operator!=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
constexpr bool operator<(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
constexpr bool operator<=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
```

```
constexpr bool operator>(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
constexpr bool operator>=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;
```

Regardless of whether they contain a valid pointer or null pointer, zero-length *spans* are all considered equal. This is considered a useful property when writing library code. If users wish to distinguish between a zero-length *span* with a valid pointer value and a *span* containing the null pointer, then they can do so by calling the *data()* member function and examining the pointer value directly.

Creating sub-spans

span offers convenient member functions for generating a new *span* that is a reduced view over its sequence. In each case, the newly constructed *span* is returned by value from the member function. As the design requires bounds-safety, these member functions are guaranteed to either succeed and return a valid *span*, or fail with undefined behavior (e.g. calling *terminate()*) if the parameters were not within range.

```
// [span.sub], span subviews
constexpr span<element_type, dynamic_extent> first(index_type count) const;
constexpr span<element_type, dynamic_extent> last(index_type count) const;
constexpr span<element_type, dynamic_extent> subspan(index_type offset,
index_type count = dynamic_extent) const;
```

first() returns a new *span* that is limited to the first N elements of the original sequence. Conversely, *last()* returns a new *span* that is limited to the last N elements of the original sequence. *subspan()* allows an arbitrary sub-range within the sequence to be selected and returned as a new *span*.

All three member functions are overloaded in forms that accept their parameters as template parameters, rather than function parameters. These overloads are helpful for creating fixed-size *span* objects from an original input *span*, whether fixed- or dynamic-size.

```
template <ptrdiff_t Count>
constexpr span<element_type, Count> first() const;
template <ptrdiff_t Count>
constexpr span<element_type, Count> last() const;
template <ptrdiff_t Offset, ptrdiff_t Count = dynamic_extent>
constexpr span<element_type, Count> subspan() const;
```

Multidimensional *span*

span as presented here only supports a single-dimension view of a sequence. This covers the most common usage of contiguous sequences in C++. *span* has convenience (such as iterators, *first()*, *last()*, and *subspan()*) and default behaviors that make most sense in a single-dimension.

Adding support for multidimensional and noncontiguous (strided) views of data is deferred to a separate type not described here. One such candidate would be the more general *view* facility described in P0009 [5]. The interface of *span* is sufficiently compatible with that of *view*, that users should not feel

any significant discontinuity between the two. In fact, it is entirely possible to implement a *span* using *view*.

Proposed Wording Changes

The following proposed wording changes against the working draft of the standard are relative to N4567 [7]. If byte type would be available in the working draft, as proposed in P0257 [6] then the functions in section [span.objectrep] would use “byte” for the placeholder *byte*.

In these changes,

Yellow highlight is used to indicate modified text or sections.

Red highlight is used to indicate deleted text.

Green highlight is used to indicate newly added text.

17.6.1.2 Headers [headers]

2 The C++ standard library provides **54** C++ library headers, as shown in Table 14.

Table 14 – C++ library headers

<algorithm>	<fstream>	<list>	<regex>	<thread>
<array>	<functional>	<locale>	<scoped_allocator>	<tuple>
<atomic>	<future>	<map>	<set>	<type_traits>
<bitset>	<initializer_list>	<memory>		<typeindex>
<chrono>	<iomanip>	<mutex>	<sstream>	<typeinfo>
<codecvt>	<ios>	<new>	<stack>	<unordered_map>
<complex>	<iosfwd>	<numeric>	<stdexcept>	<unordered_set>
<condition_variable>	<iostream>	<ostream>	<streambuf>	<utility>
<deque>	<istream>	<queue>	<string>	<valarray>
<exception>	<iterator>	<random>	<stringstream>	<vector>
<forward_list>	<limits>	<ratio>	<system_error>	

23 Containers library [containers]

23.1 General [containers.general]

2 The following subclauses describe container requirements, and components for sequence containers, **associative containers, and views** as summarized in Table 94.

Table 94 – Containers library summary

Subclause	Header(s)
23.2 Requirements	
23.3 Sequence containers	<array> <deque> <forward_list>

	<list> <vector>
23.4 Associative containers	<map> <set>
23.5 Unordered associative containers	<unordered_map> <unordered_set>
23.6 Container adaptors	<queue> <stack>
23.7 Views	

23.7 Views [views]

23.7.1 General [views.general]

1 The header defines the view `span`. A `span` is a view over a contiguous sequence of objects, the storage of which is owned by some other object.

Header synopsis

```
namespace std {

// [views.constants], constants
constexpr ptrdiff_t dynamic_extent = -1;

// [span], class template span
template <class ValueType, ptrdiff_t Extent = dynamic_extent>
class span;

// [span.comparison], span comparison operators
template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator==(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator!=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator<(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator<=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator>(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator>=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;
```

```

// [span.objectrep], views of object representation
template <class ElementType, ptrdiff_t Extent>
    constexpr span<const char, ((Extent == dynamic_extent) ?
dynamic_extent : (sizeof(ElementType) * Extent))>
as_bytes(span<ElementType, Extent> s) noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr span<char, ((Extent == dynamic_extent) ? dynamic_extent :
(sizeof(ElementType) * Extent))> as_writable_bytes(span<ElementType,
Extent> ) noexcept;

} // namespace std

```

23.7.2 Class template `span` [views.span]

1 A `span` is a view over a contiguous sequence of objects, the storage of which is owned by some other object.

2 `ElementType` is required to be a complete object type that is not an abstract class type.

3 Throughout this section, whenever a requirement fails to be met, the result is considered undefined behavior. It may – for example – cause immediate termination via a call to `terminate()`, or cause an exception to be thrown.

3 The iterators for `span` are all random access iterators and contiguous iterators.

4 For a `span<const T>`, the `iterator` and `const_iterator` types are allowed to be synonyms.

```

namespace std {
// A view over a contiguous, single-dimension sequence of objects
template <class ElementType, ptrdiff_t Extent = dynamic_extent>
class span {
public:
    // constants and types
    using element_type = ElementType;
    using index_type = ptrdiff_t;
    using pointer = element_type*;
    using reference = element_type&;
    using iterator = /*implementation-defined*/;
    using const_iterator = /* implementation-defined */;
    using reverse_iterator = reverse_iterator<iterator>;
    using const_reverse_iterator = reverse_iterator<const_iterator>;

    constexpr static index_type extent = Extent;

    // [span.cons], span constructors, copy, assignment, and destructor
    constexpr span();
    constexpr span(nullptr_t);
    constexpr span(pointer ptr, index_type count);
    constexpr span(pointer firstElem, pointer lastElem);
    template <size_t N>
        constexpr span(element_type (&arr)[N]);

```

```

template <size_t N>
    constexpr span(array<remove_const_t<element_type>, N>& arr);
template <size_t N>
    constexpr span(const array<remove_const_t<element_type>, N>& arr);
template <class Container>
    constexpr span(Container& cont);
template <class Container>
    span(const Container&&) = delete;
constexpr span(const span& other) noexcept = default;
constexpr span(span&& other) noexcept = default;
template <class OtherElementType, ptrdiff_t OtherExtent>
    constexpr span(const span<OtherElementType, OtherExtent>& other);
template <class OtherElementType, ptrdiff_t OtherExtent>
    constexpr span(span<OtherElementType, OtherExtent>&& other);
~span() noexcept = default;
constexpr span& operator=(const span& other) noexcept = default;
constexpr span& operator=(span&& other) noexcept = default;

// [span.sub], span subviews
template <ptrdiff_t Count>
    constexpr span<element_type, Count> first() const;
template <ptrdiff_t Count>
    constexpr span<element_type, Count> last() const;
template <ptrdiff_t Offset, ptrdiff_t Count = dynamic_extent>
    constexpr span<element_type, Count> subspan() const;
constexpr span<element_type, dynamic_extent> first(index_type count)
const;
constexpr span<element_type, dynamic_extent> last(index_type count)
const;
constexpr span<element_type, dynamic_extent> subspan(index_type
offset, index_type count = dynamic_extent) const;

// [span.obs], span observers
constexpr index_type length() const noexcept;
constexpr index_type size() const noexcept;
constexpr index_type length_bytes() const noexcept;
constexpr index_type size_bytes() const noexcept;
constexpr bool empty() const noexcept;

// [span.elem], span element access
constexpr reference operator[](index_type idx) const;
constexpr reference operator()(index_type idx) const;
constexpr pointer data() const noexcept;

// [span.iter], span iterator support
iterator begin() const noexcept;
iterator end() const noexcept;

const_iterator cbegin() const noexcept;
const_iterator cend() const noexcept;

reverse_iterator rbegin() const noexcept;
reverse_iterator rend() const noexcept;

const_reverse_iterator crbegin() const noexcept;

```

```

    const_reverse_iterator crend() const noexcept;

private:
    // exposition only
    pointer data_;
    index_type size_;
};

// [span.comparison], span comparison operators
template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator==(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator!=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator<(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator<=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator>(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr bool operator>=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;

// [span.objectrep], views of object representation
template <class ElementType, ptrdiff_t Extent>
    constexpr span<const byte, ((Extent == dynamic_extent) ?
dynamic_extent : (sizeof(ElementType) * Extent))>
as_bytes(span<ElementType, Extent> s) noexcept;

template <class ElementType, ptrdiff_t Extent>
    constexpr span<byte, ((Extent == dynamic_extent) ? dynamic_extent :
(sizeof(ElementType) * Extent))> as_writable_bytes(span<ElementType,
Extent> ) noexcept;

} // namespace std

```

23.7.2.1 span constructors, copy, assignment, and destructor [span.cons]

```

constexpr span();
constexpr span(nullptr_t);

```

Requires: extent == dynamic_extent || extent == 0

Effects: Constructs an empty span.

Postconditions: `size() == 0 && data() == nullptr`

Complexity: Constant.

```
constexpr span(pointer ptr, index_type count);
```

Requires: When `ptr` is null pointer then `count` shall be 0. When `ptr` is not null pointer, then it shall point to the beginning of a valid sequence of objects of at least `count` length. `count` shall always be ≥ 0 . If `extent` is not `dynamic_extent`, then `count` shall be equal to `extent`.

Effects: Constructs a `span` that is a view over the sequence of objects pointed to by `ptr`. If `ptr` is null pointer or `count` is 0 then an empty `span` is constructed.

Postconditions: `size() == count && data() == ptr`

Complexity: Constant.

```
constexpr span(pointer firstElem, pointer lastElem);
```

Requires: `[firstElem, lastElem)` is a valid range and `distance(firstElem, lastElem)  $\geq 0$` . If `extent` is not equal to `dynamic_extent`, then `distance(firstElem, lastElem)` shall be equal to `extent`.

Effects: Constructs a `span` that is a view over the range `[firstElem, lastElem)`. If `distance(firstElem, lastElem)` then an empty `span` is constructed.

Postconditions: `size() == distance(first, last) && data() == firstElem`

Complexity: The same as for `distance(first, last)`

```
template <size_t N>
    constexpr span(element_type (&arr)[N]);
template <size_t N>
    constexpr span(array<remove_const_t<element_type>, N>& arr);
template <size_t N>
    constexpr span(const array<remove_const_t<element_type>, N>& arr);
```

Requires: If `extent` is not equal to `dynamic_extent`, then `N == extent`.

Effects: Constructs a `span` that is a view over the supplied array.

Postconditions: `size() == N && (N == 0 ? data() == nullptr : data() == addressof(arr[0]))`

Complexity: Constant

```
template <class Container>
constexpr span(Container& cont);
```

Requires: The constructor shall not participate in overload resolution unless:

- Container meets the requirements of both a contiguous container (defined in 23.2.1/13) and a sequence container (defined in 23.2.3).
- The Container implements the optional sequence container requirement of operator[] (defined in Table 100).
- Container::value_type is the same as remove_const_t<element_type>.

The constructor shall not participate in overload resolution if Container is a span or array.

If extent is not equal to dynamic_extent, then cont.size() shall be equal to extent.

Effects: Constructs a span that is a view over the sequence owned by cont.

Postconditions: size() == cont.size() && data() == addressof(cont[0])

Complexity: Constant.

```
template <class Container>
span(const Container&&) = delete;
```

Requires: The constructor shall not participate in overload resolution unless:

- Container meets the requirements of both a contiguous container (defined in 23.2.1/13) and a sequence container (defined in 23.2.3).
- The Container implements the optional sequence container requirement of operator[] (defined in Table 100).
- Container::value_type is the same as remove_const_t<element_type>.

The constructor shall not participate in overload resolution if Container is a span or array.

```
constexpr span(const span& other) noexcept = default;
constexpr span(span&& other) noexcept = default;
```

Effects: Constructs a span by copying the implementation data members of another span.

Postconditions: other.size() == size() && other.data() == data()

Complexity: Constant.

```

template <class OtherElementType, ptrdiff_t OtherExtent>
constexpr span(const span<OtherElementType, OtherExtent>& other);

template <class OtherElementType, ptrdiff_t OtherExtent>
constexpr span(span<OtherElementType, OtherExtent>&& other);

```

Requires: These constructors shall not participate in overload resolution unless trying to access `OtherElementType` through an `ElementType*` would meet the rules for well-defined object access defined in 3.10/10. If `extent` is not equal to `dynamic_extent`, then `other.size()` shall be equal to `extent`.

Effects: Constructs a `span` by copying the implementation data members of another `span`, performing suitable conversions.

Postconditions: `size() == other.size()` &&
`data() == reinterpret_cast<pointer>(other.data())`

Complexity: Constant.

```

span& operator=(const span& other) noexcept = default;
span& operator=(span&& other) noexcept = default;

```

Effects: Assigns the implementation data of one `span` into another.

Postconditions: `size() == other.size()` && `data() == other.data()`

Complexity: Constant.

23.7.2.2 `span` subviews [`span.sub`]

```

template <ptrdiff_t Count>
constexpr span<element_type, Count> first() const;

```

Requires: `Count >= 0` && `Count <= size()`

Effects: Returns a new `span` that is a view over the initial `Count` elements of the current `span`.

Returns: `span(data(), Count);`

Complexity: Constant.

```

template <ptrdiff_t Count>
constexpr span<element_type, Count> last() const;

```

Requires: `Count >= 0` && `Count <= size()`

Effects: Returns a new `span` that is a view over the final `Count` elements of the current `span`.

Returns: `span(Count == 0 ? data() : data() + (size() - Count), Count)`

Complexity: Constant.

```
template <ptrdiff_t Offset, ptrdiff_t Count = dynamic_extent>
constexpr span<element_type, Count> subspan() const;
```

Requires: `(Offset == 0 || Offset > 0 && Offset < size()) && (Count == dynamic_extent || Count >= 0 && Offset + Count <= size())`

Effects: Returns a new `span` that is a view over `Count` elements of the current `span` starting at element `Offset`. If `Count` is equal to `dynamic_extent`, then a `span` over all elements from `Offset` onwards is returned.

Returns: `span(data() + Offset, Count == dynamic_extent ? size() - Offset : Count)`

Complexity: Constant

```
constexpr span<element_type, dynamic_extent> first(index_type count)
const;
```

Requires: `count >= 0 && count <= size()`

Effects: Returns a new `span` that is a view over the initial `count` elements of the current `span`.

Returns: `span(data(), count);`

Complexity: Constant.

```
constexpr span<element_type, dynamic_extent> last(index_type count)
const;
```

Requires: `Count >= 0 && Count <= size()`

Effects: Returns a new `span` that is a view over the final `Count` elements of the current `span`.

Returns: `span(Count == 0 ? data() : data() + (size() - Count), Count)`

Complexity: Constant.

```
constexpr span<element_type, dynamic_extent> subview(index_type
offset, index_type count = dynamic_extent) const;
```

Requires: `(Offset == 0 || Offset > 0 && Offset < size()) && (Count == dynamic_extent || Count >= 0 && Offset + Count <= size())`

Effects: Returns a new `span` that is a view over `Count` elements of the current `span` starting at element `Offset`. If `Count` is equal to `dynamic_extent`, then a `span` over all elements from `Offset` onwards is returned.

Returns: `span(data() + Offset, Count == dynamic_extent ? size() - Offset : Count)`

Complexity: Constant

23.7.2.2 `span` observers [`span.obs`]

```
constexpr index_type length() const noexcept;
```

Effects: Equivalent to `size()`.

```
constexpr index_type size() const noexcept;
```

Effects: Returns the number of elements accessible through the `span`.

Returns: `>= 0`

Complexity: Constant

```
constexpr index_type length_bytes() const noexcept;
```

Effects: Equivalent to `size_bytes()`.

```
constexpr index_type size_bytes() const noexcept;
```

Effects: Returns the number of bytes used for the object representation of all elements accessible through the `span`.

Returns: `size() * sizeof(element_type)`

Complexity: Constant

```
constexpr bool empty() const noexcept;
```

Effects: Equivalent to `size() == 0`.

Returns: `size() == 0`

Complexity: Constant

23.7.2.3 `span` element access [`span.elem`]

```
constexpr reference operator[] (index_type idx) const;  
constexpr reference operator() (index_type idx) const;
```

Requires: `idx >= 0 && idx < size()`

Effects: Returns a reference to the element at position `idx`.

Returns: `*(data() + idx)`

Complexity: Constant

```
constexpr pointer data() const noexcept;
```

Effects: Returns either a pointer to the first element in the sequence accessible via the `span` or the null pointer if that was the value used to construct the `span`.

Returns: (for exposition) `data`

Complexity: Constant

23.7.2.4 `span` iterator support [`span.iterators`]

```
iterator begin() const noexcept;  
const_iterator cbegin() const noexcept;
```

Returns: An iterator referring to the first element in the `span`.

Complexity: Constant

```
iterator end() const noexcept;  
const_iterator cend() const noexcept;
```

Returns: An iterator which is the past-the-end value.

Complexity: Constant

```
reverse_iterator rbegin() const noexcept;  
const_reverse_iterator crbegin() const noexcept;
```

Returns: An iterator that is semantically equivalent to `reverse_iterator(end())`.

Complexity: Constant

```
reverse_iterator rend() const noexcept;  
const_reverse_iterator crend() const noexcept;
```

Returns: An iterator that is semantically equivalent to `reverse_iterator(begin())`.

Complexity: Constant

23.7.2.5 span comparison operators [span.comparison]

```
template <class ElementType, ptrdiff_t Extent>  
constexpr bool operator==(const span<ElementType, Extent>& l, const  
span<ElementType, Extent>& r) const noexcept;
```

Effects: Equivalent to `equal(l.begin(), l.end(), r.begin(), r.end())`.

```
template <class ElementType, ptrdiff_t Extent>  
constexpr bool operator!=(const span<ElementType, Extent>& l, const  
span<ElementType, Extent>& r) const noexcept;
```

Effects: Equivalent to `!(l == r)`.

```
template <class ElementType, ptrdiff_t Extent>  
constexpr bool operator<(const span<ElementType, Extent>& l, const  
span<ElementType, Extent>& r) const noexcept;
```

Effects: Equivalent to `lexicographical_compare(l.begin(), l.end(), r.begin(), r.end())`.

```
template <class ElementType, ptrdiff_t Extent>  
constexpr bool operator>(const span<ElementType, Extent>& l, const  
span<ElementType, Extent>& r) const noexcept;
```

Effects: Equivalent to `(r < l)`.

```
template <class ElementType, ptrdiff_t Extent>
constexpr bool operator<=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;
```

Effects: Equivalent to `!(l > r)`.

```
template <class ElementType, ptrdiff_t Extent>
constexpr bool operator>=(const span<ElementType, Extent>& l, const
span<ElementType, Extent>& r) const noexcept;
```

Effects: Equivalent to `!(l < r)`.

23.7.2.6 views of object representation [span.objectrep]

```
template <class ElementType, ptrdiff_t Extent>
constexpr span<const byte, ((Extent == dynamic_extent) ?
dynamic_extent : (sizeof(ElementType) * Extent))>
as_bytes(span<ElementType, Extent> s) noexcept;
```

Effects: Constructs a `span` over the object representation of the elements in `s`.

Returns: `{ reinterpret_cast<const byte*>(s.data()), sizeof(ElementType) * s.size() }`

```
template <class ElementType, ptrdiff_t Extent>
constexpr span<byte, ((Extent == dynamic_extent) ? dynamic_extent
: (sizeof(ElementType) * Extent))> as_writable_bytes(span<ElementType,
Extent> s) noexcept;
```

Requires: This function will not participate in overload resolution when `is_const<ElementType>::true` exists.

Effects: Constructs a `span` over the object representation of the elements in `s`.

Returns: `{ reinterpret_cast<byte*>(s.data()), sizeof(ElementType) * s.size() }`

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